What one prototype actually needs.

One XL r0 rig, including adjustable suspension and purchased equipment. Most lines are unquoted planning allowances. This is not a retail chair price or an animal-ready build quote.

DIY cash, with printer & tools

$3,012.12

With paid assembly

$3,362.12

With development allowances

$5,312.12

The hardware in the chair.

Amounts are for the complete purchase line, not per piece. Exact interfaces and supplier approval are still pending. Metal parts remain metal; plastic copies are unloaded layout gauges.

Item / BOM coverage	Buy quantity & status	Low	Base	High
Outer aluminum tube stock M01, M02, M04	3 m, 25 x 25 x 2 mm 2,252.5 mm net cut length. Exact metric 6061-T6 stock, kerf and delivery need a quote. Imperial tube is not a drop-in replacement.	$45.00	$65.00	$100.00
Inner aluminum tube stock M03, M05	1 m, 20 x 20 x 2 mm 622.5 mm net cut length. Actual telescoping and corner radii must be checked.	$20.00	$30.00	$45.00
External frame finish and masking Additional allowance	1 small-batch finish operation Candidate matte forest coating on exterior outer tubes only. Mask sliding sections, tube interiors, bores, bearing interfaces and clamped joint faces. Exact process, adhesion, corrosion behavior and quote are pending. Separate from metal cutting/deburring and finish-hole work.	$35.00	$70.00	$140.00
Metal cheeks and fabrication M06, M07, M08	16 plates: 4 fixed, 4 arm, 8 corner 6 mm plate stock, cutting, deburring and finish-hole work. Includes outsourced tube drilling/match-fitting allowance; no drill press purchase assumed. Shop RFQ required.	$120.00	$220.00	$400.00
Machined pins and axles S02, S06, S07, S11, S12, W02	10 retained pivot/stop/lock/eye pins, 2 axles Headed shafts, finish fits and cross-holes are custom work. Drawing approval and actual quote pending.	$100.00	$200.00	$400.00
Metal spacers and crush sleeves S04, S05, S08, S09, S10, W03, W04, H05, H06, H07a, H07b	38 spacers and 26 sleeves Small-quantity material and machining. These are metal compression/load-path parts, not prints.	$80.00	$150.00	$300.00
Bronze pivot bushes S03	4, nominal 10 ID / 14 OD / 6 long Supplier press/running fits pending.	$12.00	$24.00	$48.00
Frame fastener sets H01, H02, H03, H04, H08, H09	26 bolts, 52 washers, 26 locknuts M6 nominal. Lengths and grades require the final joint drawing.	$25.00	$40.00	$65.00
Pin washers and retainers H10, H11	12 end washers and 12 cotters Positive retention, not friction caps. Final pin-end ligaments and cotter fit pending.	$10.00	$18.00	$30.00

All hardware/fabrication lines on this page need actual quotes.

The running gear and first spool.

Requested dimensions describe an RFQ interface. Similar-looking catalog parts are not approved substitutes. The spool is charged once, not again by estimated grams.

Item / coverage	Full quantity & status	Low	Base	High
Wheel, tire and bearing assemblies W01	2, requested 400 mm OD / 12 mm bore / 40 mm hub RFQ only. A nominal 16-inch wheel is not automatically compatible or load rated for this rig.	$90.00	$140.00	$220.00
Adjustable spring/damper assemblies S01	2 matched units 165 extended / 38 stroke / 155 installed / 8 pin / 20 eye width / 40 maximum body, all mm. No approved purchasable SKU or spring tune yet. This allowance can be exceeded.	$120.00	$220.00	$400.00
Padded rear support T01	1 professionally selected harness Capacity, anatomy and attachment compatibility need fitting and supplier review.	$80.00	$130.00	$220.00
Front harness and paired tethers T02	1 fitted set Supplier-rated softgoods; not a homemade printed saddle.	$35.00	$60.00	$100.00
Support anchor interface reserve Additional allowance	1 interface set, design unresolved Additional allowance beyond the current BOM, because the harness anchor load path is not yet designed. Not an approved part or purchase instruction.	$30.00	$60.00	$120.00
Initial PETG purchase P-tube_cap, P-inner_cap, P-index_collar, P-pivot_guard, P-drill_jig, P-coupon, P-shock_dummy	1 full 1 kg spool, 1.75 mm Cash purchase, not a measured print mass. Covers service/fixture prints and some fit-check iterations only as far as the actual slicer yield allows. Metal-part plastic copies are unloaded gauges. Do not charge consumed grams again.	$20.00	$25.00	$35.00
Printer and dryer electricity Additional allowance	One printing/iteration allowance Unmeasured allowance. Log actual hours, kWh and utility tariff; no print time is represented as observed.	$1.00	$3.00	$8.00

Read the totals with their assumptions.

Tax is an illustrative **9.75% reserve** on goods, equipment and fabrication, excluding electricity and shipping. It is not an address-specific tax calculation or a statement about labor taxability. Confirm at checkout with the CDTFA lookup. Contingency is 20% of hardware + the spool + shipping. It is not a tested failure rate.

If you already own every suitable equipment/fixture item, the same base model is **$2,030.97**. A printer alone accounts for $549 plus about $53.53 of the illustrative tax reserve. The printer is not a per-chair recurring charge.

The shop quote must include metric tube supply, plate material, hole finishes, spacers, retention and any assembly fixtures. Ask shock and wheel suppliers to approve the actual interface drawing and intended use. No shock spring rate, compatible SKU or harness-anchor design is released by this budget.

Buy once. Account for the whole purchase.

The first rig includes the entire printer, tool setup and one spool purchase. Filament grams are not charged again. Print mass, failed prints and electricity have not been measured.

Item	Quantity & evidence	Low	Base	High
Bambu Lab P2S, printer only	1, no AMS or laser bundle Observed listed price. Seller lists printer-only at $549. Verify configuration, current stock, freight and checkout price. The 256 mm bed accommodates the current nominal 220 mm part envelope; check brim and exclusions.	$549.00	$549.00	$549.00
Filament dryer	1 single-spool dryer Planning allowance, not a quote. SUNLU S2 is a capability reference, not a verified price. Match the filament maker drying guidance.	$40.00	$60.00	$90.00
Digital caliper for printed/metal parts	1, 150 mm range Observed listed price. Harbor Freight item 63711 regular listed price, no membership coupon assumed. Not a tool to squeeze or measure Buddy body width.	$24.99	$24.99	$24.99
Hand tools and measuring supplies	Wrenches, clamps, deburring tools, tape, ruler, boards Planning allowance, not a quote. Supplier fabricates metalwork. Borrowing an already suitable tool reduces cash outlay; calibration/torque procedure still needs approval.	$60.00	$100.00	$180.00
Cleanup supplies and spare nozzle	1 starter set Planning allowance, not a quote. Check printer-specific consumables. No extra build plate assumed.	$20.00	$35.00	$60.00
Dry storage and desiccant	1 starter set Planning allowance, not a quote. Reusable storage, not additional filament.	$15.00	$25.00	$40.00
Empty-rig catch/cradle and scales	1 bench fixture allowance Planning allowance, not a quote. Testing equipment, not part of the wearable chair. This is not a qualified test program or calibrated lab quote.	$50.00	$100.00	$180.00

The complete DIY cash scenario.

Cost group	Low	Base	High
Device hardware	$802.00	$1,427.00	$2,588.00
Full spool + power	$21.00	$28.00	$43.00
Printer, tools + fixture	$758.99	$893.99	$1,123.99
Shipping allowance	$60.00	$120.00	$240.00
Illustrative tax reserve	$154.15	$228.73	$365.33
20% build contingency	$176.40	$314.40	$572.60
DIY cash total	**$1,972.54**	**$3,012.12**	**$4,932.92**

Time and review are real costs.

Additional allowance	Basis	Low	Base	High
Paid assembly, if used	6 h x $25 / 10 h x $35 / 16 h x $60. Illustrative labor, not a quote. DIY cash scenarios exclude this line, but your time is not free economically.	$150.00	$350.00	$960.00
Mechanical engineering review	6 h x $100 / 8 h x $150 / 12 h x $180. Unquoted planning allowance, not a scope or sign-off commitment.	$600.00	$1,200.00	$2,160.00
Veterinary/rehabilitation fitting	Unquoted visit allowance. Buddy care continues independently; treatment and ongoing care are additional.	$150.00	$250.00	$450.00
Bench testing and additional specimens	Unquoted development allowance beyond the reusable cradle. Actual test protocol, repetitions and failures can cost more.	$250.00	$500.00	$1,000.00
DIY cash + paid assembly + development		**$3,122.54**	**$5,312.12**	**$9,502.92**

> **This does not buy a safety approval.** Review scope and test results are unresolved. More iterations, custom shocks, guarding, harness-anchor redesign, durability studies or failed specimens can exceed every allowance. Veterinary treatment and ongoing care, product liability/insurance, business overhead and production certification are not included.

What is verified, and what still needs a quote.

1. Additive Plus P2S listing: $549 printer-only configuration, listed November 8, 2026. Confirm the exact configuration, stock, freight and checkout price. No AMS assumed.
2. Harbor Freight 63711: $24.99 regular price. No membership or coupon assumed. These calipers are for parts, not Buddy's body.
3. SUNLU S2 manufacturer specification and Polymaker PETG product information are capability references. Their current prices/configurations were not verified for this model. Dryer and filament are allowances.
4. All metalwork, pins, wheels, shocks, softgoods, labor and professional services require quotes. The shock and harness-anchor interfaces do not yet have approved purchasable parts.

Before committing money.

1. Finish Buddy's measurement worksheet and clinical suitability review. Do not map groin height directly to CAD rail height.
2. Have the structural drawings, shock/wheel interfaces and harness anchors reviewed. Obtain itemized quotes against the exact configured revision, including material, finish fits, shipping and tax.
3. Slice the actual nonstructural meshes on the selected printer. Confirm orientation, build envelope, grams, supports and failed-print allowance. Do not shrink parts to fit a smaller bed.
4. Approve an empty-rig test protocol before adding ballast or an animal. Price the resulting revisions; a printed part or a costly printer does not establish capacity.

The storefront's $264.50 XL price is a future retail planning target. Its $155 production COGS envelope is unquoted and excludes this one-off custom suspension development. Neither figure is the build cost above, and this r0 package has not demonstrated that retail target.